

Sabbir Mollah

GRADUATE RESEARCH ASSISTANT | IAI, UCF

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Summary

PhD student at UCF working on multimodal unified models, memory mechanisms, and multi-agent systems. Experienced in ML research, NLP, document understanding, and production ML systems. Currently working as a **GRA** at **IAI, UCF**.

Publications

ECCV 2026

Sabbir Mollah, Rohit Gupta, Sirnam Swetha, Qingyang Liu, Ahnaf Munir, Mubarak Shah, "THE TELEPHONE GAME: EVALUATING SEMANTIC DRIFT IN UNIFIED MODELS," Accepted in ECCV 2026. Available at: <https://arxiv.org/pdf/2509.04438>

ICML 2026

Kevin Zhai, **Sabbir Mollah** (Co-First Author), Zhenyi Wang, Mubarak Shah, "CORE: CONTEXT-ROBUST REMASKING FOR DIFFUSION LANGUAGE MODELS," Accepted in ICML 2026. Available at: <https://arxiv.org/pdf/2602.04096>

ICLR (Tiny Paper) 2023

Sabbir Mollah, Rakib Mohammed, Mashrur Wasek, AKM Shahariar Azad Rabby, Fuad Rahman, Nabeel Mohammed, "Automated Mapping of Healthcare Concepts to a Standardized Healthcare Taxonomy," Available at: <https://openreview.net/forum?id=87oCobKKS6x>.

ICPR 2022

Md. Ismail Hossain, Mohammed Rakib, **Sabbir Mollah**, Fuad Rahman, and Nabeel Mohammed, "LILA-BOTI: Leveraging Isolated Letter Accumulations By Ordering Teacher Insights for Bangla Handwriting Recognition," Available at: <https://arxiv.org/abs/2205.11420>

Education

University of Central Florida

PHD IN COMPUTER SCIENCE

- Graduate Research Assistant at Institute of Artificial Intelligence | UCF IAI (PhD Advisor: **Dr. Mubarak Shah**)
- Improving multimodal model performance and memory mechanisms for multi-agent systems.

Orlando, Florida, USA

2024 - Present

North South University

B.S. IN COMPUTER SCIENCE AND ENGINEERING

- Graduated with **Magna Cum Laude distinction**, CGPA - **3.73/4.00**.
- Thesis : **Domain Adaptation** on RawNet to train language invariant speaker recognition model.

Dhaka, Bangladesh

2017 - 2021

Experience

Machine Learning Engineer

BKASH LIMITED (MOBILE FINANCIAL SERVICES - MFS)

- Extracted **information** from printed documents and predicted **missing address tokens**, e.g., post office.
- Built a graph-based **fraud prevention system** using graph database.
- Designed an English-to-Bangla **machine transliteration** system.
- Built a **loan repayment risk model** in Julia for loan risk assessment.
- Implemented backend features using **Reactive Programming** for human agent-assisted services.

Dhaka, Bangladesh

DEC 2022 - JUL 2024

Software Engineer, Machine Learning

APURBA TECHNOLOGIES LIMITED

- Built Bangla Screen Reader, an **accessibility application** for visually impaired users.
- Developed an NLP solution to map electronic health records to **SNOMED Clinical Terminology**.
- Created a Bangla inter-encoding conversion library in Python, Java, C#, and JavaScript.

Dhaka, Bangladesh

SEP 2022 - DEC 2022

Research Assistant

APURBA-NSU R&D LAB

- Improved CRNN performance on minor classes using **Knowledge Distillation**.
- Trained Mask R-CNN models to detect segments of Bangla documents.
- Worked on Bangla **speech recognition**.

Dhaka, Bangladesh

SEP 2021 - AUG 2022

Lab Instructor

ECE DEPARTMENT, NORTH SOUTH UNIVERSITY

- Taught undergraduate students **Database Management Systems using SQL** and **OOP using Java**.

Dhaka, Bangladesh

Spring 2022 & Summer 2022

Skills

Machine Learning	PyTorch, TensorFlow, Hugging Face, scikit-learn, pandas, Polars, PySpark, matplotlib
Programming	Python, Julia, Java, C, C++, C#, Haskell
Frameworks	Flask, FastAPI, Spring Boot Reactive, .NET Framework
Databases	MySQL, DynamoDB, MongoDB, Neo4j
Version Control	Git, GitHub, GitLab
Languages	Bangla, English, Italian

Volunteering Experience

NSU ACM Student Chapter

Dhaka, Bangladesh

CHAIR

2017 - 2021

- Elected as the **Chapter Chair** of this community of over 300 members.
- Planned and **organized HackNSU Season 2**, an inter-university Hackathon with more than 25 participating teams.

NSU Communications Club

Dhaka, Bangladesh

SUB EXECUTIVE

2017 - 2019

- Led the promotions team responsible for digital advertising.

NSU Cultural Club

Dhaka, Bangladesh

GENERAL MEMBER

2017 - 2019

- Helped the various artists in organizing on-campus cultural events.
- Decorated campus during national and local festivities.

Workshops

Dhaka, Bangladesh

INSTRUCTOR

2020, 2021, 2022

- Led four workshops on Python, PyTorch, and deep learning.

Atomic BAFSK Science Club

Dhaka, Bangladesh

FOUNDING GENERAL SECRETARY

2014 - 2016

- Founded the science club in secondary school.

Achievements & Awards

Robi Datathon 2.0

Robi, Axiata Ltd., Dhaka

FINALISTS

2022

- Implemented cost-sensitive learning to improve machine learning model performance.

MIST Inter-University ICT Innovation Fest Hackathon

MIST, Dhaka

CHAMPIONS

2021

- Designed and deployed a handwriting recognition model.

IOT For Tomorrow

NSU IEEE SB, Dhaka

FIRST RUNNERS-UP

2019

- Presented a fire detection circuit using a microcontroller integrated with Firebase.

Electrathon

NSU IEEE SB, Dhaka

2ND RUNNERS-UP

2018

- Solved various hardware and software related problems.